

Aerodynamics group project report

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The following report documents some validation strategies for three well-known aerodynamic numerical tools for potential flows, namely the Hess–Smith panel method, XFOIL, and the Weissinger vortex-lattice method, along with a critical discussion of the results obtained for specific study cases based on available data for the Canadair CL-415.

1. Introductory remarks

All three solvers leverage linearity to construct complex flow fields from the superposition of simple solutions of the underlying Laplace equation. Consequently, several hypotheses common to all three methods must be assumed: incompressible and irrotational flow, with negligible viscous effects and volume forces. XFOIL also includes a simplified viscous model. It is crucial to keep these hypotheses in mind, as they guide the interpretation and validity of our results. Reliable predictions cannot be expected outside the conditions under which these assumptions hold, regardless of any validation achieved by comparing different methods. Each solver also relies on specific additional assumptions, for instance regarding geometry, which will be discussed when relevant

2. Hess-Smith method

A first validation strategy is to compare results from similar numerical methods. For the completed MATLAB implementation of the Hess-Smith method, the results should match those produced by XFOIL for the same airfoil at the same angle of attack (1°). The two methods are similar in the sense that they both are panel methods relying on slightly different elementary solutions. However, XFOIL also models the effect of the viscous boundary layer on the potential flow. The value of $C_l = 0.1177$ obtained by applying the Kutta-Joukowski theorem compares well to the value of $C_l = 0.1170$ obtained from XFOIL. Similarly, the moment coefficient obtained with Hess-Smith in MATLAB is $C_m = -6.5758 \times 10^{-4}$ and from XFOIL $C_m = -0.0008$, using the quarter-chord point as the moment reference point. The airfoil panel discretization was obtained from the AirfoilTools database to avoid numerical oscillations observed near the trailing edge when imposing a closed trailing edge in XFOIL and increasing panel density in Hess-Smith. The C_p distributions of the two methods closely match as shown in Fig.1. All comparisons hold for conditions in which the boundary layer and wake are thin, and no separation occurs, as viscous effects, which are not considered in any form in Hess-Smith, can be neglected. Accounting for a viscous layer allows for the examination of laminar-to-turbulent transition and separation along the airfoil surface, as seen in the next section.

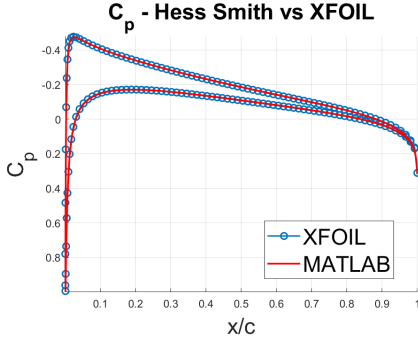


FIGURE 1. C_p distribution comparison Hess-Smith code vs XFOIL.

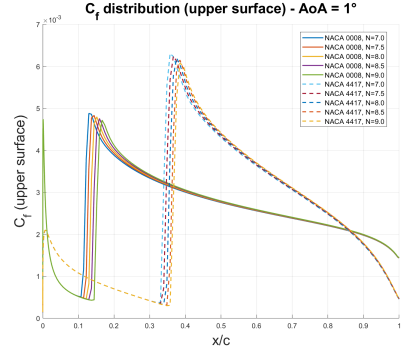


FIGURE 2. C_f distribution vs N_{crit} for NACA 0008 and NACA 4417.

3. Effect of N_{crit} on transition and separation for two airfoils

In this paragraph, the behavior of the symmetric airfoil NACA 0008 is compared with that of NACA 4417. In particular, the laminar-to-turbulent transition of the boundary layer was analyzed along with its possible separation. The analysis was performed using XFOIL, assuming compatible flight conditions with the chosen airfoils and varying the flow turbulence level represented by the N_{crit} parameter, which represents the critical amplification number of perturbations according to the transition model based on the linear stability theory e^n [Djeddi (2022)].

The NACA 4417 airfoil was chosen as it represents a compatible configuration with the mission profile characteristics of the Canadair CL-415, which can feature both high angles of attack and low flight speed.

3.1. Simulation Method

The simulation parameters of the analysis were chosen to represent flight conditions of the Canadair CL-415. Assuming an airspeed of 300 km/h at sea level, Reynolds number was set to $Re = 2 \times 10^7$, computed using the formula $Re = \frac{Vc}{\nu}$, where V is the aircraft velocity, c is the wing's chord of 3.53 m and ν is the standard air's kinematic viscosity. The flow turbulence level was varied through the N_{crit} parameter, with values chosen between 9 and 7 to avoid excessively low levels of perturbations in order to remain within the hypotheses of the e^n theory. Simulations were performed at an angle of attack of 1° , as to remain within the pre-stall regime and starting assumptions for both airfoils. In XFOIL, the number of iterations was increased: an iterative approach was adopted since the program uses the results of the previous iteration as the initial condition for the next. This approach improves numerical stability and ensures reliable results' convergence.

3.2. Results

Simulations show that if N_{crit} decreases, an earlier laminar-to-turbulent transition occurs for both airfoils. Transition is identified in Fig.2 by an upstream increase in the skin-friction coefficient. A direct comparison between the two airfoils can be seen in the same figure where it can be observed that the transition on the NACA 4417 airfoil occurs further downstream compared to the NACA 0008. Overall, the results confirm the strong influence of the free-stream turbulence level on where the boundary-layer transition occurs, as modeled through the N_{crit} parameter. The delayed transition observed on the NACA 4417 highlights the role of airfoil geometry, and in particular of airfoil camber and thickness, in reducing the adverse pressure gradient and stabilizing the

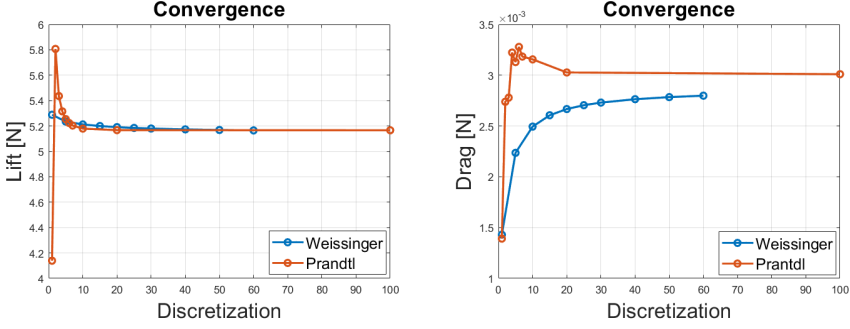


FIGURE 3. Convergence of Prandtl and Weissinger methods for a rectangular 3D wing.

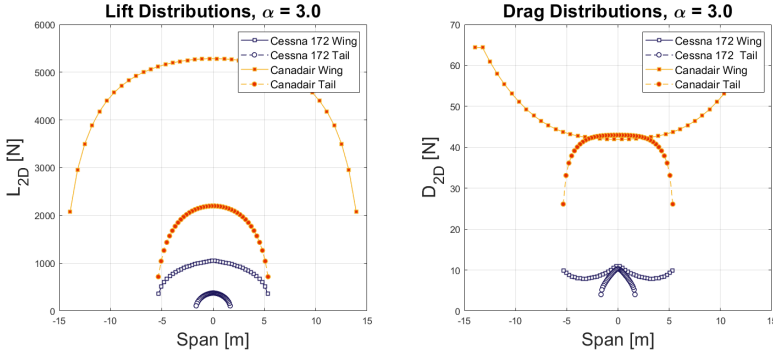


FIGURE 4. Lift and Drag distributions of both planes' surfaces.

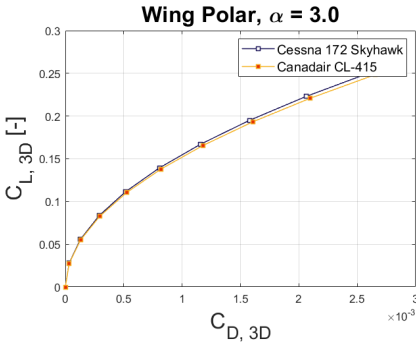


FIGURE 5. Wing aerodynamic polars.

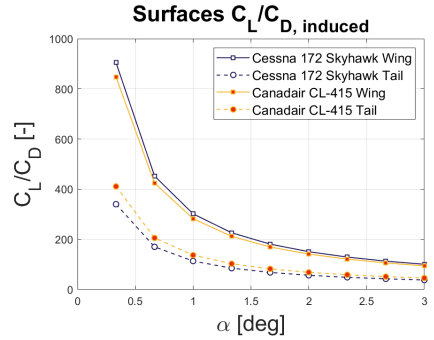


FIGURE 6. Lift to induced Drag Ratio.

boundary layer with respect to a thin symmetric airfoil within the considered operating conditions.

4. Weissinger method

Another common validation strategy for a model is a direct comparison with analytical predictions. Despite the lack of closed form solutions for a 3-dimensional wing, meaningful results can still be obtained and used for comparison. For example, given the simple case of a straight wing with high aspect ratio, Prandtl's lifting line theory can be applied. The Prandtl-Lanchester equation is solved numerically for the circulation distribution Γ , allowing for the evaluation of L and D of the entire wing. These values can then be

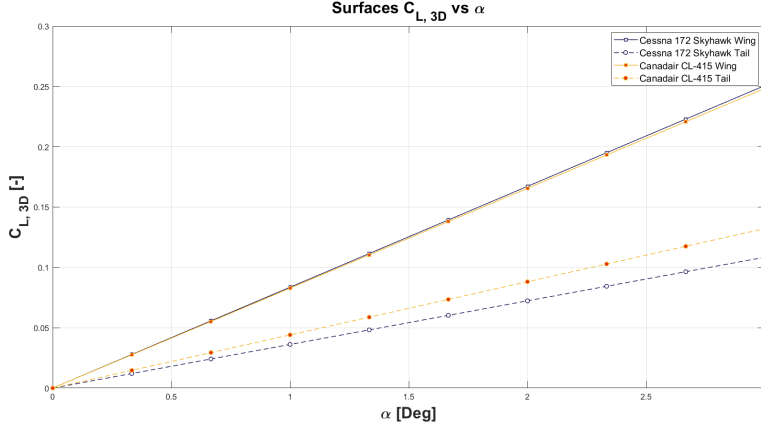


FIGURE 7. $C_L - \alpha$ of wing and tail for both planes.

compared to those obtained from the completed code of Weissinger’s method for the same wing geometry, dimensions, and conditions. It is crucial to note that the resulting lift and drag converge when increasing the order of approximation of Γ of the first method and the wings’ panel mesh resolution of the second, as seen in Fig. 3.

4.1. Analysis and comparison of the Cessna 172 and the Canadair CL-415

A two-surface modeling approach was adopted for both planes. The geometric characteristics and dimensions of the wing and the horizontal tail were estimated from data provided by McIver (2003) and Avincis (2024).

The lift distribution appears smoother for a greater aspect ratio. This becomes clear in Fig. 4, when comparing the wing lift distributions of the two aircraft, the Canadair’s appears flatter in the wing’s center. From the Drag distributions, it is possible to observe stronger relative vortex influence at the tips of the smaller wing. The aerodynamic polars of Fig. 5 have a reasonable parabolic trend, which further corroborates the method. Fig. 6 shows the $C_L/C_{D_{ind}}$ ratios: for the same C_L , the C172 wing generates less induced drag than the CL-415. The opposite trend is seen for the tails. Finally, it is interesting to comment on the $C_L - \alpha$ plots seen in Fig. 7. The aspect ratio, taper ratio, and dihedral angle all influence the slope of the curve. Having very similar values for all three, the wings’ curves are akin. A similar plot should be expected for the tails for the same reason, but this is not the case. A possible explanation lies in the tail positioning relative to the wing. The Cessna has the second surface significantly closer to the wing wake and is therefore more strongly influenced by it, which explains the different $C_L - \alpha$ slope and the worse $C_L/C_{D_{ind}}$ when compared to that of the Canadair’s tail.

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